TITLE PAGE

SIH26106 / SOFTWARE / BLOCKCHAIN & CYBERSECURITY

MailTrace

Email threat detection + origin forensics

Upload a saved email. See the evidence behind its risk, its likely infrastructure, and its relationship to other cases.

PROBLEM STATEMENT ID	SIH26106
PROBLEM STATEMENT TITLE	AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform
THEME	Blockchain & Cybersecurity
PS CATEGORY	Software
TEAM ID	**[ENTER PORTAL TEAM ID]**
TEAM NAME	**[ENTER REGISTERED TEAM NAME]**

From one suspicious message to a connected case file.

MailTrace combines detection, origin context and campaign correlation in one analyst workflow.

PROPOSED SOLUTION

01
CAPTURE
.eml bytes stay local; hash the exact evidence.

02
EXTRACT
From, Reply-To, Received, URLs and MIME parts.

03
EXPLAIN
Auth results, risk score and reasons an analyst can cite.

04
CONNECT
Shared Reply-To, domain or hop IP becomes a campaign lead.

PROTOTYPE OUTPUT

CAMPAIGN CORRELATION

shared indicators -> campaign candidate

07 / CLOUD HOPS
mail-secure.net
Reply-To + 18.184.10.20 / AWS

Reply-To + hop
+ domain

08 / CAMPAIGN TWIN
ecure.net
+ 18.184.10.20 / AWS

Detection = reasons | Origin = hosting intelligence | Correlation = a lead, not an identity claim

SMART INDIA HACKATHON 2026

TEAM NAME

TECHNICAL APPROACH

Evidence pipeline: small, explainable modules.

The working prototype is deterministic first; NLP is one supporting signal, not the whole product.

01 .EML
raw bytes + SHA-256

02 MIME
headers, body, URLs, attachments

03 AUTH
SPF / DKIM / DMARC + alignment

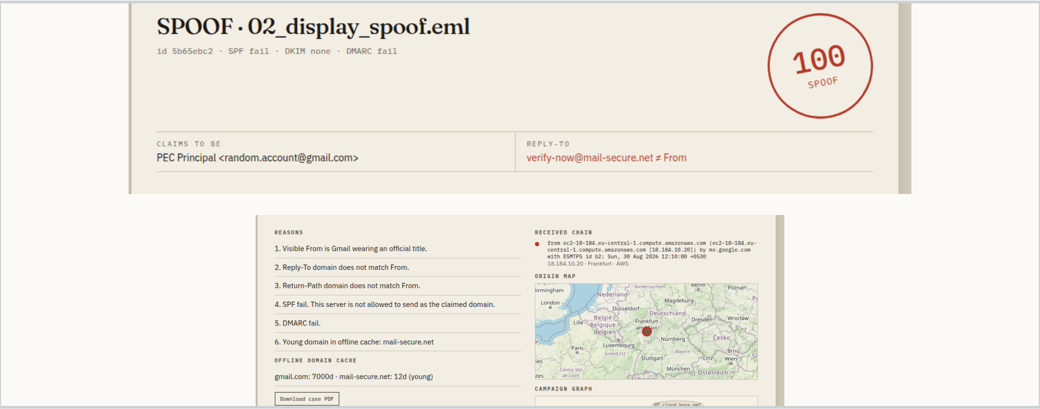
04 ORIGIN
Received chain + hosting intel

05 OUTPUT
score, map, graph, PDF

IMPLEMENTED PROTOTYPE

- FastAPI + Python email parser + SQLite case store
- NetworkX campaign graph + Leaflet hop map + ReportLab forensic PDF
- Offline demo intelligence keeps the laptop demo reproducible and safe

REAL LOCALHOST PROOF



Captured from the local case screen: score, reasons, hop map and evidence footer.

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FEASIBILITY AND VIABILITY

Feasible now; harden the edges as adoption grows.

The first version runs on a laptop and avoids fragile live dependencies. The production path is modular.

NOW / IDEA-ROUND PROTOTYPE

- saved .eml upload; no mailbox credentials
- 8 crafted cases for clean, spoof, phish, BEC and campaign paths
- offline demo geo and domain cache for reproducible runs
- local SQLite case store, graph and PDF evidence

NEXT / PRODUCTION HARDENING

- live DNS and cryptographic auth verification
- maintained GeoIP / ASN data and organisation deployment
- access control, PII masking and retention policies
- larger labelled corpus, calibration and analyst feedback

RISKS AND MITIGATIONS

RISK	MITIGATION IN MAILTRACE
Headers can be missing or forged	show confidence, preserve raw evidence, never call a lead proof
IP geolocation is not a human location	report hosting city / ISP only; explicitly reject GPS attribution
Email data is sensitive	local-first workflow; no Gmail ingest; future masking, access and retention controls

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IMPACT AND BENEFITS

Impact: turn a vague warning into an actionable case.

MailTrace is for people who must explain what happened, preserve evidence and decide what to do next.

THE ANALYST DIFFERENCE

FILTER

spam / not spam

A label without context leaves the next question unanswered.

MAILTRACE CASE FILE

SPF fail · Reply-To mismatch · earliest hop

Frankfurt / AWS · related case 08 · SHA-256

reasons + origin context + campaign lead

TARGET USERS AND VALUE

IT / helpdesk	triage a suspicious message with reasons they can cite
SOC / incident response	preserve original evidence and connect related messages
universities / SMEs	run locally without handing an inbox to a third party

PILOT MEASURES / NO INVENTED NUMBERS

time-to-triage · evidence completeness · linked-campaign discovery · analyst agreement

RESEARCH AND REFERENCES

Research basis and implementation references.

The prototype maps directly to the problem statement while keeping attribution and privacy claims bounded.

PROBLEM + STANDARDS

SIH26106	Official PS: NLP/ML detection, header forensics, IP geo, domain intelligence, graph correlation and forensic reporting.
RFC 5322	Internet Message Format: structured headers, message IDs and Received fields.
RFC 7208 / 6376 / 7489	SPF, DKIM and DMARC concepts used for authentication evidence and alignment.

BUILD REFERENCES

Python email	standard-library MIME parsing; attachments are named and hashed, never executed.
NetworkX + Leaflet	relationship graph and hop map for analyst inspection.
ReportLab + SHA-256	portable case PDF and integrity fingerprint of the exact uploaded bytes.
SpamAssassin corpus	optional TF-IDF baseline only; it is not presented as origin forensics.

SCOPE AND SAFETY

- Local crafted .eml files only for the demo; no Gmail/Outlook production ingest and no real phishing messages.
- Geo output is hosting infrastructure with confidence, never the exact location of a person.
- The evidence hash is an integrity fingerprint, not a public blockchain or cryptocurrency ledger.

Source links: sih.gov.in/sih2026PS · rfc-editor.org · docs.python.org · networkx.org · leafletjs.com